

APPLIED STATISTICS
(for AIML Students)
B. Tech 3rd semester. 2025-2029 Batch
Syllabus

UNIT-1

The Role of Statistics in Engineering:

The Engineering Method and Statistical Thinking – Data Collection

Engineering Data and Basic Principles:

Retrospective Studies – Observational Studies – Designed Experiments – Observing Processes Over Time – Mechanistic and Empirical Models.

Data Description and Representation:

Collection of Data – Classification and Tabulation of Data – Stem-and-Leaf Diagrams – Frequency Distributions and Histograms – Box Plots – Time Sequence Plots – Probability Plots.

UNIT-2

Descriptive Statistics:

Measures of Central Tendency – Measures of Dispersion – Skewness and Kurtosis.

Correlation and Regression:

Scatter Diagram – Types of Correlation – Karl Pearson's Coefficient of Correlation – Spearman's Rank Correlation – Method of Least Squares – Linear

UNIT-3

Sampling:

Types of Sampling – Sampling Distributions – Sampling Distribution of the Mean.

Point Estimation of Parameters:

General Concepts of Point Estimation – Unbiased Estimators – Variance of a Point Estimator – Standard Error – Methods of Point Estimation (Method of Moments, Method of Maximum Likelihood).

Statistical Intervals for a Single Sample:

- (i) Confidence Interval for the Mean of a Normal Distribution (Variance Known)
- (ii) Confidence Interval for the Mean of a Normal Distribution (Variance Unknown)
- (iii) Confidence Interval for the Variance and Standard Deviation of a Normal Distribution
- (iv) Large-Sample Confidence Interval for a Population Proportion.

UNIT-4

Tests of Hypotheses for a Single Sample:

Tests of Statistical Hypotheses – General Procedure for Hypothesis Testing

- (i) Test on the Mean of a Normal Distribution (Variance Known)
- (ii) Test on the Mean of a Normal Distribution (Variance Unknown)
- (iii) Test on the Variance and Standard Deviation of a Normal Distribution.

UNIT-5

Statistical Inference for Two Samples:

- (i) Inference for the Difference in Means of Two Normal Distributions (Variances Known)
- (ii) Inference for the Difference in Means of Two Normal Distributions (Variances Unknown)
- (iii) Inference on the Variances of Two Normal Distributions
- (iv) Inference on Two Population Proportions.

Reference Books:

01. Douglas C. Montgomery and George C. Runger. Applied Statistics and Probability for Engineers, (3rdEdn), John Wiley and Sons, Inc., New York, 2003.
02. Robert H. Carver and Jane Gradwohl Nash. Doing Data Analysis with SPSS Version 18.0, (Indian Edition) Cengage Learning, New Delhi, 2012.
03. Richard A. Johnson and C.B. Gupta, Probability and Statistics for Engineers, (7thEdn.), Pearson Education, Indian Impression 2006.

UNIT–III

(Inferential Statistics)

APPLIED STATISTICS

(for AIML Students)

Unit-3 (Inferential Statistics)

Topic: Sampling

B. Tech 3rd semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

Introduction

Statistics may be broadly studied under two main headings:

1. Descriptive Statistics
2. Inductive Statistics

Descriptive Statistics

Descriptive Statistics deals with the collection, organization, presentation, and summarization of data. It describes the important features of a given set of data using tables, graphs, and numerical measures.

Examples: Mean, Median, Mode, Range, Variance, Standard Deviation, Tables, Bar Charts, Histograms, etc.

Inductive Statistics

Inductive Statistics, also called **Inferential Statistics**, deals with drawing conclusions or making predictions about a population based on information obtained from a sample.

It includes estimation, hypothesis testing, correlation, regression, and prediction.

Inductive Statistics or Inferential Statistics

The most important terminologies in the study of **Inferential Statistics** are

Population

A **Population** is the complete set of individuals, objects, or observations about which a statistical study is concerned.

Example: If we want to study the marks of all students in a university, the marks of all students constitute the **population**.

Sample

A **Sample** is a subset of the population selected for the purpose of statistical analysis. A properly selected sample should represent the characteristics of the population as closely as possible.

Example: If 200 students are selected from a university of 5000 students for a study, the 200 selected students constitute the **sample**.

Sampling

In statistical studies, it is often difficult or impossible to collect information from every member of a population. Therefore, a smaller group called a **sample** is selected from the population and studied.

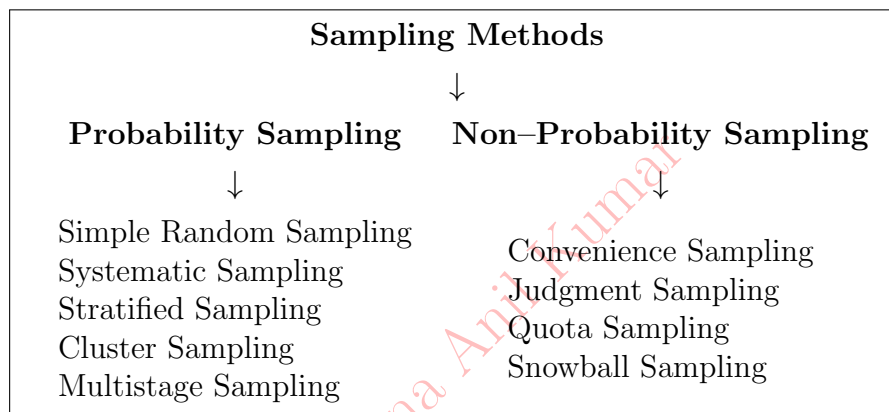
Need for Sampling

Sampling is useful because:

1. It reduces the cost of data collection.
2. It saves time.
3. It requires fewer resources.
4. It is useful when the population is very large.
5. It may provide sufficiently accurate information about the population.
6. It is sometimes the only practical method when testing destroys the item being tested.

Types of Sampling

Sampling methods can broadly be classified into two categories:



Comparison of Sampling Methods

Method	Main Idea	Example
Simple Random	Every member has an equal chance of selection	Selecting students using lottery method
Systematic	Select every k th member	Selecting every 10th student
Stratified	Divide population into strata and sample from each	Sampling students from each branch
Cluster	Divide population into clusters and select clusters	Selecting some classes and surveying them
Multistage	Sampling is performed in several stages	State → University → College → Student
Convenience	Select easily available members	Surveying students present in class
Judgment	Selection based on researcher judgment	Selecting experienced engineers
Quota	Select a fixed number from different groups	Selecting 20 students from each branch
Snowball	Participants identify additional participants	Surveying a hard-to-reach population

Population Parameter and Sample Statistic

A numerical characteristic of a population is called a **parameter**.

A numerical characteristic calculated from a sample is called a **statistic**.

Characteristic	Population	Sample
Mean	μ	$\bar{X}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\hat{p}$
Skewness	β_1	$b_1(B_1)$
Kurtosis	β_2	$b_2(B_2)$
Moments	μ_r	m_r
Correlation coefficient	ρ	r

Statistic as a function:

The sample statistics are functions of the sample observations and vary from sample to sample. Thus, if t is any general statistic, then we may write t as a function of the sample observations $x_1, x_2, \dots, x_n$ as given below:

$$t = t(x_1, x_2, \dots, x_n)$$

Number of Possible Samples

Let a population consists of N units and a sample of size n is selected.

The number of possible **samples without replacement** is

$${}^N C_n = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

For example, if $N = 5$ and $n = 2$,

$$\binom{5}{2} = \frac{5!}{2!3!} = 10.$$

Hence, there are 10 possible samples.

Sampling Distribution

The probability distribution of a statistic obtained from all possible random samples of a fixed size drawn from a population is called the **sampling distribution of that statistic**.

Sampling Distribution of the Mean

The **sampling distribution of the mean** is the probability distribution of the sample mean $\bar{X}$ obtained from all possible samples of a fixed size n from a population.

Example:

A population consists of the four values

2, 4, 6, 8.

The number of possible ordered **samples with replacement** is

$$N^n$$

For example, if $N = 5$ and $n = 2$,

$$5^2 = 25.$$

Hence, there are 25 possible samples.

All possible samples of size $n = 2$ are selected without replacement. Find the sampling distribution of the sample mean $\bar{X}$.

Answer:

The number of possible samples of size 2 from a population of 4 values is

$$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \boxed{6}.$$

The six possible samples are

Sample	Sample Values	$\bar{X}$
1	(2,4)	3
2	(2,6)	4
3	(2,8)	5
4	(4,6)	5
5	(4,8)	6
6	(6,8)	7

Sampling Distribution of $\bar{X}$

Since all six samples are equally likely,

$\bar{X}$	Frequency	$P(\bar{X})$
3	1	$\frac{1}{6}$
4	1	$\frac{1}{6}$
5	2	$\frac{2}{6}$
6	1	$\frac{1}{6}$
7	1	$\frac{1}{6}$
Total	6	1

Hence, the sampling distribution of $\bar{X}$ is

$$P(\bar{X} = 3) = \frac{1}{6}, \quad P(\bar{X} = 4) = \frac{1}{6}, \quad P(\bar{X} = 5) = \frac{2}{6}, \quad P(\bar{X} = 6) = \frac{1}{6}, \quad P(\bar{X} = 7) = \frac{1}{6}$$

Standard Error in Statistic

The standard deviation of the sampling distribution of a statistic is known as **Standard Error (S.E.)**. Thus, the Standard Error of the statistic t is given by

$$S.E.(t) = \sqrt{Var(t)} = \sqrt{\frac{1}{k} \sum_{i=1}^k (t_i - \bar{t})^2}$$

Here,

- $k = \binom{N}{n} = \frac{N!}{n!(N-n)!}$ is the possible number of samples.
- $t_1, t_2, \dots, t_k$ are the values of statistic for k possible samples.
- $\bar{t}$ is the mean of the statistic values.

The standard errors of the sampling distributions of some of the well known statistics are presented in the table as below.

Statistic	Notation	Standard Error
Sample Mean	$\bar{X}$	$\frac{\sigma}{\sqrt{n}}$
Sample Proportion	p	$\sqrt{\frac{PQ}{n}}$
Sample Standard Deviation	s	$\frac{\sigma^2}{\sqrt{2n}}$
Sample Variance	s^2	$\sigma^2 \sqrt{\frac{2}{n}}$
Sample Median		$1.25331 \frac{\sigma}{\sqrt{n}}$
Sample Correlation Coefficient	r	$\frac{1 - \rho^2}{\sqrt{n}}$
Difference of Two Means	$\bar{x}_1 - \bar{x}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
Difference of Two Proportions	$p_1 - p_2$	$\sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$
Difference of Two sample s.d.	$s_1 - s_2$	$\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$

Important Notes

- Here n is the sample size.
- P is the population proportion and $Q = 1 - P$.
- σ^2 is the population variance and σ is the population standard deviation.
- For practical applications, when σ is unknown, it is usually replaced by the sample standard deviation s .
- For sampling without replacement from a finite population of size N , the standard error of the

sample mean becomes $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$.

APPLIED STATISTICS

(for AIML Students)

Unit-3 (Inferential Statistics)

Topic: Theory of Estimation

B. Tech 3rd semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

Classification of Statistical Inference

In statistical inference, the primary objective is to obtain information about an unknown population parameter based on information obtained from a sample.

Statistical inference provides various techniques for generalizing the results obtained from a sample to the entire population. These techniques also help us assess the validity and reliability of such generalizations and quantify the degree of confidence associated with the conclusions drawn.

Based on the above discussion, the techniques of statistical inference are broadly classified into the following two major areas:

- (i) **Theory of Estimation**
- (ii) **Testing of Hypothesis**

Theory of Estimation

The process of using sample information to estimate an unknown population parameter is called **estimation**.

There are two main types of estimation:

1. **Point Estimation**
2. **Interval Estimation**

Point Estimation

A **point estimate** is a single numerical value used to estimate an unknown population parameter.

Interval Estimation

In **interval estimation**, an interval or range of values is constructed from sample data within which the unknown population parameter is expected to lie with a specified level of confidence.

Estimator

An **estimator** is a function of sample observations used to estimate an unknown population parameter.

Example :

In a population, a random sample of size n is selected:

$$X_1, X_2, \dots, X_n.$$

The **sample mean** is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

and is used as a **point estimator** of the unknown population mean μ .

Once a particular sample is observed, say

$$x_1, x_2, \dots, x_n,$$

the corresponding sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Here,

- $\bar{X}$ is the **estimator**; it is a random variable.
- $\bar{x}$ is the **estimate**; it is the numerical value obtained from the observed sample.
- μ is the unknown population parameter being estimated.

Good Estimator

A good estimator is one which is as close to the true value of the parameter as possible. The following are some of the criteria which should be satisfied by a good estimator.

1. **Unbiasedness**
2. **Consistency**
3. **Efficiency**
4. **Sufficiency**

Unbiasedness

A statistic

$$t = t(X_1, X_2, \dots, X_n)$$

is an unbiased estimate of the corresponding population parameter θ if its expected value is equal to the parameter being estimated. That is,

$$E(t) = \theta.$$

the mean value of sampling distribution of the statistic is equal to the parameter.

Example:

The sample mean is an unbiased estimator of the population mean. That is,

$$E(\bar{X}) = \mu.$$

Interpretation:

Suppose a population has mean μ , and let $X_1, X_2, \dots, X_n$ denote the observations in a random sample.

Each observation has expectation μ :

$$E(X_i) = \mu, \quad i = 1, 2, \dots, n.$$

The sample mean is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Taking expectation on both sides,

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

By the **linearity of expectation**,

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i).$$

Since

$$E(X_1) = E(X_2) = \cdots = E(X_n) = \mu,$$

we obtain

$$E(\bar{X}) = \frac{1}{n}(n\mu) = \mu.$$

Therefore,

$$\boxed{E(\bar{X}) = \mu}.$$

Hence, the sample mean $\bar{X}$ is an **unbiased estimator** of the population mean μ .

Example :

A population consists of the four values

$$2, 4, 6, 8.$$

All possible samples of size $n = 2$ are selected with replacement. Find the sampling distribution of the sample mean $\bar{X}$. Also, verify the unbiasedness of sample mean estimator.

Solution

The population is

$$\{2, 4, 6, 8\}.$$

Since sampling is done **with replacement**, each observation in the sample can take any of the four population values.

Therefore, the total number of possible samples of size $n = 2$ is

$$4^2 = 16.$$

The sample mean is given by

$$\boxed{\bar{X} = \frac{X_1 + X_2}{2}}.$$

Sampling Distribution

List all possible samples

Sample	(X_1, X_2)	$\bar{X}$	Sample	(X_1, X_2)	$\bar{X}$
1	(2, 2)	2	9	(6, 2)	4
2	(2, 4)	3	10	(6, 4)	5
3	(2, 6)	4	11	(6, 6)	6
4	(2, 8)	5	12	(6, 8)	7
5	(4, 2)	3	13	(8, 2)	5
6	(4, 4)	4	14	(8, 4)	6
7	(4, 6)	5	15	(8, 6)	7
8	(4, 8)	6	16	(8, 8)	8

The possible values of $\bar{X}$ and their frequencies are:

$\bar{X}$	Frequency	$P(\bar{X})$
2	1	$\frac{1}{16}$
3	2	$\frac{2}{16}$
4	3	$\frac{3}{16}$
5	4	$\frac{4}{16}$
6	3	$\frac{3}{16}$
7	2	$\frac{2}{16}$
8	1	$\frac{1}{16}$
Total	16	1

Population Mean

The population mean is

$$\mu = \frac{2 + 4 + 6 + 8}{4} = 5.$$

Mean of Sampling Distribution

The mean of the sampling distribution is

$$E(\bar{X}) = \sum \bar{X} P(\bar{X}).$$

Therefore,

$$\begin{aligned} E(\bar{X}) &= 2 \left(\frac{1}{16} \right) + 3 \left(\frac{2}{16} \right) + 4 \left(\frac{3}{16} \right) + 5 \left(\frac{4}{16} \right) + 6 \left(\frac{3}{16} \right) + 7 \left(\frac{2}{16} \right) + 8 \left(\frac{1}{16} \right) \\ &= \frac{2 + 6 + 12 + 20 + 18 + 14 + 8}{16} = \frac{80}{16} = 5. \end{aligned}$$

Alternatively:

The mean of the sampling distribution is

$$\begin{aligned} &\frac{1}{16} (2 + 3 + 3 + 4 + 4 + 4 + 5 + 5 + 5 + 5 + 6 + 6 + 6 + 7 + 7 + 8) \\ &= 5. \end{aligned}$$

Thus,

$$E(\bar{X}) = 5 = \mu.$$

Conclusion

The mean of the sampling distribution of $\bar{X}$ is

$$E(\bar{X}) = 5.$$

Since the population mean is

$$\mu = 5,$$

we have

$$\boxed{E(\bar{X}) = \mu}.$$

Therefore, the sample mean $\bar{X}$ is an **unbiased estimator** of the population mean μ .

Example:

If t is an unbiased estimator of the population mean μ , where

$$t = \frac{2X_1 + X_2 + \lambda X_3}{5},$$

find the value of λ .

Solution:

Since t is an unbiased estimator of the population mean μ ,

$$E(t) = \mu.$$

Assuming X_1, X_2, X_3 are observations from the same population with mean μ ,

$$E(X_1) = E(X_2) = E(X_3) = \mu.$$

Now,

$$E(t) = E\left(\frac{2X_1 + X_2 + \lambda X_3}{5}\right).$$

Using the linearity of expectation,

$$E(t) = \frac{2E(X_1) + E(X_2) + \lambda E(X_3)}{5}.$$

Therefore,

$$E(t) = \frac{2\mu + \mu + \lambda\mu}{5} = \frac{(3 + \lambda)\mu}{5}.$$

Since t is unbiased,

$$\frac{(3 + \lambda)\mu}{5} = \mu.$$

For $\mu \neq 0$,

$$\frac{3 + \lambda}{5} = 1.$$

Hence,

$$3 + \lambda = 5,$$

and therefore,

$$\boxed{\lambda = 2}.$$

Example:

The Sample Variance

$$s^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

is not an unbiased estimator of population variance σ^2 . However,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

provides an unbiased estimator of population variance σ^2 . Thus,

$$E(s^2) \neq \sigma^2 \quad \text{and} \quad E(S^2) = \sigma^2$$

Example 1: Sample Variance with Denominator n is a Biased Estimator

Consider the population

$$\{2, 4, 6\}.$$

Find the sampling distribution of variance with sample size 2. Also, show that sample variance is biased estimator for population variance.

Solution:

The population mean is

$$\mu = \frac{2 + 4 + 6}{3} = 4.$$

Therefore, the population variance is

$$\sigma^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{4+0+4}{3} = \frac{8}{3}.$$

Now, samples of size $n = 2$ are selected with replacement. There are

$$3^2 = 9$$

possible ordered samples.

Consider the variance estimator

$$s_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

For $n = 2$,

$$s_n^2 = \frac{1}{2} \sum_{i=1}^2 (X_i - \bar{X})^2.$$

The sampling distribution of s_n^2 is obtained as follows:

Sample	Sample Mean $\bar{X}$	s_n^2
(2, 2)	2	0
(2, 4)	3	1
(2, 6)	4	4
(4, 2)	3	1
(4, 4)	4	0
(4, 6)	5	1
(6, 2)	4	4
(6, 4)	5	1
(6, 6)	6	0

Hence, the expectation of the estimator is

$$\begin{aligned} E(s_n^2) &= \frac{0 + 1 + 4 + 1 + 0 + 1 + 4 + 1 + 0}{9} \\ &= \frac{12}{9} \\ &= \frac{4}{3}. \end{aligned}$$

But the population variance is

$$\sigma^2 = \frac{8}{3}.$$

Thus,

$$E(s_n^2) = \frac{4}{3} \neq \frac{8}{3} = \sigma^2.$$

Therefore,

$$\boxed{E(s_n^2) \neq \sigma^2}$$

and hence

$$\boxed{s_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is a biased estimator of } \sigma^2.}$$

In fact,

$$E(s_n^2) = \frac{n-1}{n} \sigma^2,$$

which shows that this estimator systematically underestimates the population variance.

Example 2: Sample Variance with Denominator $n - 1$ is an Unbiased Estimator

Consider the same population

$$\{2, 4, 6\}.$$

Find the sampling distribution of variance with sample size 2. Also, show that sample variance with the formula

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

is unbiased estimator for population variance.

Solution

The population mean is

$$\mu = \frac{2 + 4 + 6}{3} = 4.$$

The population variance is

$$\sigma^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}.$$

Again, samples of size $n = 2$ are selected with replacement.

Consider the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

For $n = 2$,

$$S^2 = \sum_{i=1}^2 (X_i - \bar{X})^2.$$

The values of S^2 for all possible samples are:

Sample	Sample Mean $\bar{X}$	S^2
(2, 2)	2	0
(2, 4)	3	2
(2, 6)	4	8
(4, 2)	3	2
(4, 4)	4	0
(4, 6)	5	2
(6, 2)	4	8
(6, 4)	5	2
(6, 6)	6	0

Therefore,

$$\begin{aligned}
 E(S^2) &= \frac{0 + 2 + 8 + 2 + 0 + 2 + 8 + 2 + 0}{9} \\
 &= \frac{24}{9} \\
 &= \frac{8}{3}.
 \end{aligned}$$

Since

$$\sigma^2 = \frac{8}{3},$$

we have

$$E(S^2) = \sigma^2.$$

Therefore,

$$E(S^2) = \sigma^2$$

and hence

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is an unbiased estimator of } \sigma^2.$$

Consistency

A statistic $t = t_n = t(x_1, x_2, \dots, x_n)$ based on a sample of size n is said to be **consistent** estimator of the parameter θ if it converges in probability to θ . i. e. if $t_n \rightarrow \theta$ as $n \rightarrow \infty$. Symbolically,

$$\lim_{n \rightarrow \infty} P(t_n \rightarrow \theta) = 1$$

An equivalent condition is:

$$E(t_n) \rightarrow \theta \quad \text{and} \quad \text{Var}(t_n) \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty$$

Examples:

For any distribution,

- (i) sample mean $\bar{X}$ is a consistent estimator of population mean μ
- (ii) sample proportion p is a consistent estimator of population proportion P
- (iii) sample variance s^2 is a consistency estimator of population variance σ^2

Example:

Let T_1 and T_2 be consistent estimators of μ_1 and μ_2 respectively. Prove that

$$aT_1 + bT_2$$

is a consistent estimator of

$$a\mu_1 + b\mu_2.$$

Solution

Given that T_1 is consistent estimator for μ_1 , and T_2 is consistent estimator for μ_2 .

That is, under the mean–variance consistency criterion,

$$E(T_1) \rightarrow \mu_1, \quad E(T_2) \rightarrow \mu_2,$$

and

$$\text{Var}(T_1) \rightarrow 0, \quad \text{Var}(T_2) \rightarrow 0.$$

Let

$$T = aT_1 + bT_2.$$

Using the linearity of expectation,

$$E(T) = E(aT_1 + bT_2).$$

$$E(T) = aE(T_1) + bE(T_2).$$

$$E(T) \rightarrow a\mu_1 + b\mu_2.$$

Thus,

$$\boxed{E(T) \rightarrow a\mu_1 + b\mu_2}$$

Assuming that T_1 and T_2 are uncorrelated,

$$\text{Var}(T) = \text{Var}(aT_1 + bT_2).$$

Therefore,

$$\text{Var}(T) = a^2 \text{Var}(T_1) + b^2 \text{Var}(T_2).$$

Since

$$\text{Var}(T_1) \rightarrow 0, \quad \text{Var}(T_2) \rightarrow 0,$$

we obtain

$$\text{Var}(T) \rightarrow 0.$$

Thus,

$$\boxed{\text{Var}(T) \rightarrow 0}$$

Conclusion:

We have shown that

$$E(T) \rightarrow a\mu_1 + b\mu_2$$

and

$$\text{Var}(T) \rightarrow 0.$$

for

$$\boxed{T = aT_1 + bT_2}.$$

Therefore, by the consistency criterion,

$$\boxed{aT_1 + bT_2 \text{ is a consistent estimator of } a\mu_1 + b\mu_2.}$$

Efficiency

In practice, more than one unbiased or consistent estimator may exist for the same population parameter. In such cases, **efficiency** provides a criterion for comparing estimators and selecting the one with greater precision.

Suppose T_1 and T_2 are two unbiased estimators of the same parameter θ . Among unbiased estimators, the estimator with the **smaller variance** is said to be more efficient.

That is, if

$$E(T_1) = E(T_2) = \theta$$

and

$$\text{Var}(T_1) < \text{Var}(T_2),$$

then T_1 is more efficient than T_2 .

Therefore,

Among unbiased estimators, smaller variance implies greater efficiency.

Example :

For a normal population $N(\mu, \sigma^2)$, the sample mean and sample median are unbiased and consistent estimators of the population mean μ . Also,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

and

$$\text{Var}(Md) \approx \frac{1.57\sigma^2}{n}$$

Thus, for large samples

$$\text{Var}(\bar{X}) < \text{Var}(Md)$$

That is, the estimator sample mean ($\bar{X}$) is more efficient than the estimator sample median (MD) for population mean μ .

Definition

If t is the most efficient estimator of a parameter θ with variance ν , and t_1 is any other estimator with variance ν_1 , then the efficiency E of t_1 is defined as:

$$E = \frac{\nu}{\nu_1}.$$

Since t is the most efficient estimator, its sampling variance ν is minimum, i.e.,

$$\nu \leq \nu_1 \quad \Rightarrow \quad \frac{\nu}{\nu_1} \leq 1 \quad \Rightarrow \quad E \leq 1.$$

Note:

The efficiency of any estimator can not be exceed unity.

The estimator having the smaller variance is considered more efficient.

Example :

If X_1, X_2, X_3 constitute a random sample of size 3 from a normal population with population mean μ and variance σ^2 , find the efficiency of

$$\frac{X_1 + 2X_2 + X_3}{4}$$

relative to

$$\frac{X_1 + X_2 + X_3}{3}.$$

Solution:

Since X_1, X_2, X_3 is a random sample from $N(\mu, \sigma^2)$, we have

$$\text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X_3) = \sigma^2.$$

Further, since the sample is random, X_1, X_2, X_3 are mutually independent and hence

$$\text{Var}(a_1X_1 + a_2X_2 + a_3X_3) = a_1^2 \text{Var}(X_1) + a_2^2 \text{Var}(X_2) + a_3^2 \text{Var}(X_3).$$

Let

$$T_1 = \frac{X_1 + 2X_2 + X_3}{4} \quad \text{and} \quad T_2 = \frac{X_1 + X_2 + X_3}{3}.$$

Hence, using the above result, we get

$$\begin{aligned} \text{Var}(T_1) &= \text{Var}\left(\frac{X_1 + 2X_2 + X_3}{4}\right) \\ &= \frac{1}{4^2} \text{Var}(X_1 + 2X_2 + X_3) \\ &= \frac{1}{16} [\text{Var}(X_1) + 2^2 \text{Var}(X_2) + \text{Var}(X_3)] \\ &= \frac{1}{16} (\sigma^2 + 4\sigma^2 + \sigma^2) = \frac{6\sigma^2}{16} = \boxed{\frac{3\sigma^2}{8}}. \end{aligned}$$

Similarly,

$$\begin{aligned} \text{Var}(T_2) &= \text{Var}\left(\frac{X_1 + X_2 + X_3}{3}\right) \\ &= \frac{1}{3^2} \text{Var}(X_1 + X_2 + X_3) \\ &= \frac{1}{9} [\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)] \\ &= \frac{1}{9} (\sigma^2 + \sigma^2 + \sigma^2) = \frac{3\sigma^2}{9} = \boxed{\frac{\sigma^2}{3}}. \end{aligned}$$

Therefore,

$$\text{Var}(T_2) = \frac{\sigma^2}{3} < \frac{3\sigma^2}{8} = \text{Var}(T_1).$$

That is, T_2 is more efficient than T_1 . Hence, the efficiency of T_1 relative to T_2 is given by

$$E = \frac{\text{Var}(T_2)}{\text{Var}(T_1)}.$$

Therefore,

$$E = \frac{\frac{3\sigma^2}{9}}{\frac{3\sigma^2}{8}} = \frac{8}{9} = 0.89.$$

$$\boxed{E = \frac{8}{9} \approx 0.89}$$

Example:

Suppose X_1, X_2, X_3, X_4 are independent observations from a population with mean μ and variance σ^2 .

Consider the following two estimators of μ :

$$T_1 = \frac{X_1 + X_2 + X_3 + X_4}{4} \quad \text{and} \quad T_2 = \frac{X_1 + X_2}{2}.$$

Both T_1 and T_2 are unbiased estimators of μ . Determine which estimator is more efficient. Also, find the relative efficiency of T_1 with respect to T_2 .

Solution:

Given that:

$$E(T_1) = \mu \quad \text{and} \quad E(T_2) = \mu$$

Further,

$$\begin{aligned} \text{Var}(T_1) &= \frac{1}{16} [\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \text{Var}(X_4)] & \text{Var}(T_2) &= \frac{1}{4} [\text{Var}(X_1) + \text{Var}(X_2)] \\ &= \frac{4\sigma^2}{16} & &= \frac{2\sigma^2}{4} \\ &= \frac{\sigma^2}{4}. & &= \frac{\sigma^2}{2}. \end{aligned}$$

Therefore,

$$\text{Var}(T_1) = \frac{\sigma^2}{4} < \frac{\sigma^2}{2} = \text{Var}(T_2).$$

Hence,

$$\boxed{T_1 \text{ is more efficient than } T_2.}$$

The relative efficiency of T_1 with respect to T_2 is

$$\text{RE}(T_1, T_2) = \frac{\text{Var}(T_2)}{\text{Var}(T_1)} = \frac{\sigma^2/2}{\sigma^2/4} = 2.$$

Thus, T_1 has twice the efficiency of T_2 under this definition. **Equivalently**, the variance of T_1 is one-half of the variance of T_2 .

Sufficiency

A statistic

$$t = t(x_1, x_2, \dots, x_n)$$

is sufficient for θ if it contains all the information in the sample regarding θ .

Examples:

- Sample mean $\bar{X}$ is a sufficient estimator of population mean μ .
- Sample proportion p is a sufficient estimator of population proportion P .

Properties:

- A sufficient estimator is most efficient if one exists.
- It is always consistent.
- It may or may not be unbiased.

APPLIED STATISTICS

(for AIML Students)

Unit-3 (Inferential Statistics)

Topic: Method of Maximum Likelihood

B. Tech 3rd semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

Maximum Likelihood Estimation

Maximum Likelihood Estimation (MLE) is a method of **point estimation** used to estimate an unknown population parameter from a sample.

Let

$$X_1, X_2, \dots, X_n$$

be a random sample from a population with p.d.f./p.m.f.

$$p(x, \theta),$$

where θ is unknown.

- The likelihood is the product of the probabilities/densities of the observed sample values.
- The Maximum Likelihood Estimator (MLE) of θ is obtained by maximizing the likelihood function

$$L(\theta) = \prod_{i=1}^n p(x_i, \theta)$$

Necessary and Sufficient conditions Maximum Likelihood Estimation

For maximum likelihood, the necessary and sufficient conditions are

$$\boxed{\frac{\partial L}{\partial \theta} = 0} \quad \text{and} \quad \boxed{\frac{\partial^2 L}{\partial \theta^2} < 0.}$$

Equivalently,

$$\boxed{\frac{\partial(\log L)}{\partial \theta} = 0} \quad \text{and} \quad \boxed{\frac{\partial^2(\log L)}{\partial \theta^2} < 0.}$$

Since $\log L$ is a non-decreasing function of L , the likelihood equation may be written as

$$\boxed{\frac{\partial(\log L)}{\partial \theta} = 0.}$$

Thus, in practice, the MLE is usually obtained by the following steps:

1. Construct the likelihood function $L(\theta)$.
2. Take the natural logarithm to obtain the log-likelihood $\ell(\theta) = \log L(\theta)$.
3. Differentiate $\ell(\theta)$ with respect to θ .
4. Set the first derivative equal to zero and solve for θ .
5. Verify that the solution gives a maximum.

Maximum Likelihood Estimation for Normal Distribution

Example :

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population $X_i \sim N(\mu, \sigma^2)$, $i = 1, 2, \dots, n$, where μ and σ^2 are unknown parameters.

The probability density function (p.d.f.) of X_i is

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma^2} \right], \quad i = 1, 2, \dots, n.$$

Find the maximum likelihood estimators (MLEs) of μ and σ^2 .

Solution

For the random sample of Normal population, i.e. $X_i \sim N(\mu, \sigma^2)$, the probability density function is

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma^2} \right] \quad \text{where } i = 1, 2, \dots, n.$$

The likelihood function is

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^n f(x_i; \mu, \sigma^2) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma^2} \right] \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \end{aligned}$$

Now, this likelihood function has to be maximized with respect to μ and σ^2 . Instead of maximizing $L(\mu, \sigma^2)$ directly, we maximize its logarithm.

Let

$$\ell(\mu, \sigma^2) = \log L(\mu, \sigma^2).$$

Then,

$$\ell(\mu, \sigma^2) = \log \left[\left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \right].$$

Using the properties of logarithms,

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Estimation for μ :

First order and second order derivatives of the log-likelihood with respect to μ are:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \quad \text{and} \quad \frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}.$$

For maximum likelihood estimation, set the first derivative equal to zero:

$$\frac{\partial \ell}{\partial \mu} = 0.$$

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0.$$

$$\begin{aligned}\sum_{i=1}^n (x_i - \mu) &= 0 \\ \sum_{i=1}^n x_i - \sum_{i=1}^n \mu &= 0. \\ \sum_{i=1}^n x_i - n\mu &= 0.\end{aligned}$$

Therefore,

$$\begin{aligned}n\mu &= \sum_{i=1}^n x_i. \\ \mu &= \frac{1}{n} \sum_{i=1}^n x_i.\end{aligned}$$

Further,

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2} < 0 \quad \text{as } n > 0, \sigma^2 > 0$$

Since the sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

the maximum likelihood estimator of μ is

$$\hat{\mu} = \bar{X}$$

or equivalently,

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Estimation for σ^2 :

The log-likelihood function is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Now,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2(\sigma^2)^2}$$

For maximum likelihood estimation,

$$\frac{\partial \ell}{\partial \sigma^2} = 0.$$

Thus,

$$-\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2(\sigma^2)^2} = 0.$$

Multiplying throughout by $2(\sigma^2)^2$, we obtain

$$-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 = 0.$$

Hence,

$$n\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2.$$

Therefore,

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Further,

$$\begin{aligned}\frac{\partial \ell^2}{\partial (\sigma^2)^2} &= \frac{n}{2(\sigma^2)^2} - \frac{\sum_{i=1}^n (x_i - \mu)^2}{(\sigma^2)^3} \\ &= \frac{\frac{n}{2}\sigma^2 - \sum_{i=1}^n (x_i - \mu)^2}{(\sigma^2)^3} \\ &= \frac{\frac{n}{2}\sigma^2 - n\sigma^2}{(\sigma^2)^3} < 0\end{aligned}$$

$\therefore \mu = \hat{\mu} = \bar{X}$, we have

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Hence,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Coclusion:

For a random sample

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2),$$

the maximum likelihood estimators are

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

i.e. Sample Mean is the estimator for Population Mean

and

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

i.e. Sample Variance is the estimator for Population Variance

Example : MLE of Bernoulli Parameter

Let

$$X_1, X_2, \dots, X_n$$

be a random sample from a Bernoulli distribution with parameter p .

The probability mass function is

$$P(X = x) = p^x (1 - p)^{1-x}, \quad x = 0, 1.$$

Find the maximum likelihood estimator of p .

Solution:

For observations $x_1, x_2, \dots, x_n$, the likelihood function is

$$L(p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}.$$

Therefore,

$$L(p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}.$$

Taking logarithms,

$$\ell(p) = \left(\sum x_i\right) \ln p + \left(n - \sum x_i\right) \ln(1 - p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}.$$

Setting this equal to zero,

$$\frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p} = 0.$$

Therefore,

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1 - p}.$$

Cross multiplication gives

$$\left(\sum x_i\right) (1 - p) = p \left(n - \sum x_i\right).$$

Hence,

$$\sum x_i = np.$$

Therefore,

$$\boxed{\hat{p}_{MLE} = \frac{\sum x_i}{n} = \bar{X}.}$$

Thus, the MLE of the Bernoulli parameter is the sample proportion.

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APPLIED STATISTICS

(for AIML Students)

Unit-3 (Inferential Statistics)

Topic: Interval Estimation

B. Tech 3rd semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

Interval Estimation

Interval estimation is the method of estimating an unknown population parameter by a range of values with a specified confidence level.

Confidence Interval

If t is the statistic used to estimate the population parameter θ , then, for a symmetric confidence interval, the **confidence interval** is given by

$$\left[t - S.E.(t) c_{\alpha/2}, t + S.E.(t) c_{\alpha/2} \right]$$

such that,

$$P \left[t - S.E.(t) c_{\alpha/2} \leq \theta \leq t + S.E.(t) c_{\alpha/2} \right] = 1 - \alpha.$$

Here,

- (i) $t - S.E.(t) c_{\alpha/2}$ and $t + S.E.(t) c_{\alpha/2}$ are called the **lower and upper confidence limits**, respectively.
- (ii) α is the **level of significance**.
- (iii) $1 - \alpha$ is called the **confidence level** or **confidence coefficient**.
- (iv) $c_{\alpha/2}$ is the appropriate **critical value** corresponding to the significance level α , determined from the relevant sampling distribution.

Example: Critical Values

Some commonly used two-tailed critical values for normal distribution are:

α	0.5	0.31735	0.10	0.05	0.0455	0.02	0.01	0.0027
Confidence Level	50%	68.27%	90%	95%	95.45%	98%	99%	99.73%
$z_{\alpha/2}$	0.6745	1.00	1.645	1.960	2.00	2.326	2.576	3.000

Confidence Interval for the Mean of a Normal Distribution when Variance Known

The $(1 - \alpha)100\%$ confidence interval for the population mean μ is

$$\left[\bar{X} - z_{\alpha/2} S.E.(\bar{X}), \bar{X} + z_{\alpha/2} S.E.(\bar{X}) \right].$$

For sampling with replacement:

$$S.E.(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

For sampling without replacement from a finite population:

$$S.E.(\bar{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}},$$

where $\sqrt{\frac{N-n}{N-1}}$ is the **finite population correction (FPC)**.

Example 1

The quality control manager of a tyre company has a sample of 100 tyres and has found the mean life time to be 30214 km. The population standard deviation is 860 km. Construct a 95% confidence interval for the mean life time for this particular brand of tyre.

Solution:

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Solution:

Given,

$$n = 100, \quad \bar{x} = 30214, \quad \sigma = 860$$

For a 95% confidence level,

$$(1 - \alpha) = 95\%, \quad \frac{\alpha}{2} = 0.025$$

Hence,

$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

The standard error of the mean is

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{860}{\sqrt{100}} = 86$$

The 95% confidence interval for the population mean is

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Therefore,

$$CI = [30214 - 1.96(86), 30214 + 1.96(86)]$$

$$= [30214 - 168.56, 30214 + 168.56]$$

$$CI = [30045.44, 30382.56]$$

Thus, the 95% confidence interval for the mean life time of the tyres is

$$30045.44 \text{ km} < \mu < 30382.56 \text{ km}$$

Example 2

A finite population consists of $N = 500$ engineering components. The population standard deviation is known to be $\sigma = 20$. A simple random sample of $n = 100$ components is selected **without replacement**. The sample mean lifetime is found to be $\bar{X} = 75$. Find the 95% confidence interval for the population mean μ .

Solution

Solution

Given that

$$N = 500, \quad n = 100, \quad \bar{X} = 75, \quad \sigma = 20 \quad (1 - \alpha)100\% = 95\%.$$

For a finite population, the standard error of the sample mean is

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{20}{\sqrt{100}} \sqrt{\frac{400}{499}} \approx 2(0.8959) = 1.7918.$$

Hence,

$$\boxed{SE(\bar{X}) \approx 1.792.}$$

The required confidence level is $1 - \alpha = 95\%$. That is,

$$1 - \alpha = 0.95 \quad \implies \quad \alpha = 0.05 \quad \text{and} \quad \frac{\alpha}{2} = 0.025.$$

From the standard normal distribution table,

$$\boxed{z_{\alpha/2} = z_{0.025} = 1.96.}$$

Then, the $(1 - \alpha)100\%$ confidence interval for the population mean μ is

$$\begin{aligned} CI &= [\bar{X} - z_{\alpha/2} S.E.(\bar{X}), \bar{X} + z_{\alpha/2} S.E.(\bar{X})] \\ &= [75 - (1.96) \times 1.792, 75 + (1.96) \times 1.792] \\ &= [75 - 3.512, 75 + 3.512] \\ &= [71.488, 78.512]. \end{aligned}$$

Hence,

$$\boxed{71.488 \leq \mu \leq 78.512.}$$

We are 95% confident that the population mean μ lies between

$$\boxed{71.488 \text{ and } 78.512}.$$

Confidence Interval for the Mean of a Normal Distribution when Variance Unknown

The $(1 - \alpha)100\%$ confidence interval for the population mean μ is

$$\left[\bar{X} - t_{\alpha/2, n-1} S.E.(\bar{X}), \bar{X} + t_{\alpha/2, n-1} S.E.(\bar{X}) \right].$$

where

- $n - 1 =$ degrees of freedom,
- $t_{\alpha/2, n-1} =$ critical value from the t -distribution.

In case of sampling with replacement from large population

$$S.E.(\bar{X}) = \frac{S}{\sqrt{n}}$$

In case of sampling without replacement from a finite population

$$S.E.(\bar{X}) = \frac{S}{\sqrt{n}} \frac{N - n}{N - 1}$$

where

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}.$$

Remark: In case of sample size n is sufficiently large, the critical value $t_{\alpha/2, n-1}$ is replaced by $z_{\alpha/2}$.

Examples : σ^2 is unknown

Example 3

A random sample of 10 observations is taken from a normal population. The sample mean is 52 and the sample standard deviation is 4. Find the 95% confidence interval for the population mean μ .

Solution

Given that

$$n = 10, \quad \bar{X} = 52, \quad S = 4.$$

Also,

$$1 - \alpha = 0.95 \implies \alpha = 0.05 \implies \frac{\alpha}{2} = 0.025.$$

and the degrees of freedom are

$$df = n - 1 = 10 - 1 = 9.$$

From the t -distribution table,

$$t_{0.025, 9} = 2.262.$$

Since the population variance is unknown, the CI for μ is

$$CI = \bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

$$CI = 52 \pm 2.262 \frac{4}{\sqrt{10}}.$$

$$CI = 52 \pm 2.862.$$

Hence,

$$49.138 \leq \mu \leq 54.862$$

Therefore, the 95% confidence interval for the population mean is

$$(49.138, 54.862).$$

That is, we are 95% confident that the population mean μ lies between

$$49.138 \text{ and } 54.862.$$

Example 4

In a random sample of 64 of 600 road crossings in a town, the mean number of automobile accidents per year was found to be 4.2 and the sample S.D. was 0.8. Construct a 95% confidence interval for the mean number of automobile accidents per crossing per year.

Solution:

Given,

$$n = 64, \quad N = 600, \quad \bar{x} = 4.2, \quad S = 0.8$$

For a 95% confidence level,

$$(1 - \alpha) = 95\%, \quad Z_{\alpha/2} = 1.96$$

The standard error of the mean is

$$\begin{aligned} SE(\bar{x}) &= \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \\ &= \frac{0.8}{\sqrt{64}} \sqrt{\frac{600-64}{600-1}} \\ &= \frac{0.8}{8} \sqrt{\frac{536}{599}} \\ &\approx 0.095 \end{aligned}$$

Therefore, the 95% confidence interval is

$$\begin{aligned} CI &= [\bar{x} - Z_{\alpha/2} SE(\bar{x}), \bar{x} + Z_{\alpha/2} SE(\bar{x})] \\ &= [4.2 - (1.96)(0.095), 4.2 + (1.96)(0.095)] \\ &= [4.013, 4.386] \end{aligned}$$

Hence, the 95% confidence interval for the mean number of automobile accidents per crossing per year is

$$\boxed{4.013 < \mu < 4.386}$$

Confidence Interval for a Population Proportion.

The population proportion is given by

$$P = \frac{\text{Number of population elements possessing the characteristic}}{\text{Total number of elements in the population}}.$$

and

$$Q = 1 - P,$$

The sample proportion is

$$p = \frac{\text{Number of sample elements possessing the characteristic}}{\text{Total number of elements in the sample}}.$$

and

$$q = 1 - p,$$

Then, the $(1 - \alpha)100\%$ confidence interval for the population proportion P is

$$\boxed{\left[p - z_{\alpha/2} S.E.(p), p + z_{\alpha/2} S.E.(p) \right]}.$$

The standard error formula in different scenarios as follows:

Statistic	Standard Error	Remark	Population Size
p	$SE(p) = \sqrt{\frac{PQ}{n}}$	P is known. Sampling with replacement	Large
p	$SE(p) = \sqrt{\frac{pq}{n}}$	P is unknown. Sampling with replacement	Large
p	$SE(p) = \sqrt{\frac{PQ}{n}} \sqrt{\frac{N-n}{N-1}}$	P is known. Sampling without replacement	Finite
p	$SE(p) = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$	P is unknown. Sampling without replacement	Finite

Example 5

Out of 300 households in a town, 123 have TV sets. Find the 95% confidence limits for the true value of the proportion of households with TV sets in the whole town.

Solution

Given,

$$n = 300, \quad p = \frac{123}{300} = 0.41$$

$$q = 1 - p = 1 - 0.41 = 0.59$$

The standard error of the proportion is

$$SE(p) = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.41(0.59)}{300}} = 0.0284$$

For a 95% confidence level,

$$(1 - \alpha) \times 100\% = 95\%, \quad z_{\alpha/2} = 1.96$$

Therefore, the 95% confidence interval is

$$CI = p \pm z_{\alpha/2} SE(p)$$

$$= 0.41 \pm 1.96(0.0284)$$

$$= 0.41 \pm 0.0557$$

Hence,

$$CI = (0.3543, 0.4657)$$

Thus, the 95% confidence limits for the true proportion of households having TV sets are approximately

$$0.3543 < P < 0.4657$$

Example 6

Out of 20,000 cases, a sample of 600 ledger accounts was taken to test the accuracy of posting and balancing accounts, wherein 45 mistakes were found. Within which limits can the proportion of defective cases be expected at 5% level?

Solution

Given,

$$n = 600, \quad p = \frac{45}{600} = 0.075$$

$$q = 1 - p = 1 - 0.075 = 0.925$$

The standard error of the proportion is

$$SE(p) = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.075(0.925)}{600}} = 0.0108$$

At 5% level of significance, the confidence level is

$$(1 - \alpha) \times 100\% = 95\%, \quad z_{\alpha/2} = 1.96$$

Therefore, the 95% confidence interval is

$$CI = p \pm z_{\alpha/2}SE(p)$$

$$= 0.075 \pm 1.96(0.0108)$$

$$= 0.075 \pm 0.0212$$

Hence,

$$CI = (0.0538, 0.0962)$$

Thus, the proportion of defective cases in the population can be expected to lie between approximately

$$0.0538 < P < 0.0962$$

Since the total number of cases is 20,000, the corresponding number of defective cases will lie between

$$0.0538 \times 20,000 = 1076$$

and

$$0.0962 \times 20,000 = 1924.$$

Therefore, the number of defective cases can be expected to lie between approximately

$$[1076 \text{ and } 1924].$$

Example 7:

A finite population consists of $N = 1000$ elements, of which 40% possess a particular characteristic. A simple random sample of $n = 100$ elements is selected without replacement. Find the 95% confidence interval for the population proportion P if the observed sample proportion is $p = 0.43$.

Solution.

Given,

$$N = 1000, \quad n = 100,$$

and the population proportion is known:

$$P = 0.40 \quad \implies Q = 1 - P = 0.60.$$

Since the population is finite and sampling is without replacement, the standard error is

$$SE(p) = \sqrt{\frac{PQ}{n}} \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{(0.40)(0.60)}{100}} \sqrt{\frac{1000-100}{1000-1}} = 0.04649.$$

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

The observed sample proportion is

$$p = 0.43.$$

Therefore,

$$CI = p \pm z_{\alpha/2} SE(p) = 0.43 \pm 1.96(0.04649) = 0.43 \pm 0.09112.$$

Hence,

$$\boxed{0.33888 < P < 0.52112}$$

Example 8:

A finite population contains $N = 2000$ elements. A simple random sample of $n = 200$ elements is selected without replacement. Among the sampled elements, 84 possess a particular characteristic. Find the 95% confidence interval for the population proportion P .

Solution.

Given,

$$N = 2000, \quad n = 200, \quad x = 84.$$

The sample proportion is

$$p = \frac{x}{n} = \frac{84}{200} = 0.42.$$

Therefore,

$$q = 1 - p = 1 - 0.42 = 0.58.$$

Since the population is finite, sampling is without replacement, and P is unknown, the standard error is

$$SE(p) = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{(0.42)(0.58)}{200}} \sqrt{\frac{2000-200}{2000-1}} = 0.03312.$$

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

Thus,

$$CI = p \pm z_{\alpha/2} SE(p) = 0.42 \pm 1.96(0.03312) = 0.42 \pm 0.06492.$$

Therefore,

$$\boxed{0.35508 < P < 0.48492}$$

Hence, the 95% confidence interval for the population proportion is approximately 35.5% to 48.5%.

Some more practice problems

Example 1

A random sample of 64 observations is selected from a normal population. The sample mean is 52 and the population standard deviation is 8. Find the 95% confidence interval for the population mean.

Solution:

Given

$$n = 64, \quad \bar{x} = 52, \quad \sigma = 8.$$

For a 95% confidence level,

$$\alpha = 0.05, \quad z_{\alpha/2} = z_{0.025} = 1.96.$$

The confidence interval is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Therefore,

$$52 \pm 1.96 \left(\frac{8}{\sqrt{64}} \right).$$

Since

$$\frac{8}{8} = 1,$$

we obtain

$$52 \pm 1.96.$$

Hence,

$$50.04 < \mu < 53.96.$$

Thus, the 95% confidence interval for the population mean is

$$(50.04, 53.96).$$

Example 2: Large Population, Sampling with Replacement, P Known

A large population contains 40% of elements possessing a particular characteristic. A random sample of 400 elements is selected with replacement. Find the 95% confidence interval for the population proportion P .

Solution.

Given,

$$P = 0.40, \quad Q = 1 - P = 0.60, \quad n = 400.$$

Since the population is large, sampling is with replacement, and P is known, the standard error of the sample proportion is

$$SE(p) = \sqrt{\frac{PQ}{n}}.$$

Therefore,

$$SE(p) = \sqrt{\frac{(0.40)(0.60)}{400}} = \sqrt{0.0006} = 0.024495.$$

For a 95% confidence level,

$$z_{\alpha/2} = z_{0.025} = 1.96.$$

Suppose that the observed sample proportion is

$$p = 0.42.$$

Hence, the confidence interval is

$$p \pm z_{\alpha/2} \text{SE}(p).$$

Thus,

$$0.42 \pm 1.96(0.024495).$$

The margin of error is

$$1.96(0.024495) = 0.04801.$$

Therefore,

$$\boxed{0.37199 < P < 0.46801}$$

or approximately,

$$\boxed{(0.372, 0.468)}.$$

Hence, the 95% confidence interval for the population proportion is approximately 37.2% to 46.8%.

Example 3: Large Population, Sampling with Replacement, P Unknown

In a large population, a random sample of 400 elements is selected with replacement. Among them, 168 elements possess a particular characteristic. Find the 95% confidence interval for the population proportion P .

Solution.

The sample size is

$$n = 400.$$

The number of elements possessing the characteristic is

$$x = 168.$$

Therefore, the sample proportion is

$$p = \frac{x}{n} = \frac{168}{400} = 0.42.$$

Hence,

$$q = 1 - p = 1 - 0.42 = 0.58.$$

Since P is unknown, we estimate the standard error using the sample proportion:

$$\text{SE}(p) = \sqrt{\frac{pq}{n}}.$$

Therefore,

$$\text{SE}(p) = \sqrt{\frac{(0.42)(0.58)}{400}} = \sqrt{0.000609} = 0.02468.$$

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

Thus,

$$p \pm z_{\alpha/2} \text{SE}(p)$$

gives

$$0.42 \pm 1.96(0.02468).$$

The margin of error is

$$1.96(0.02468) = 0.04837.$$

Therefore,

$$0.37163 < P < 0.46837$$

or approximately,

$$(0.372, 0.468).$$

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